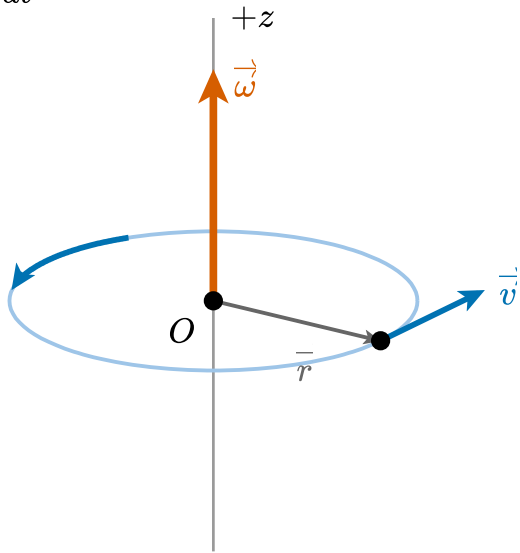
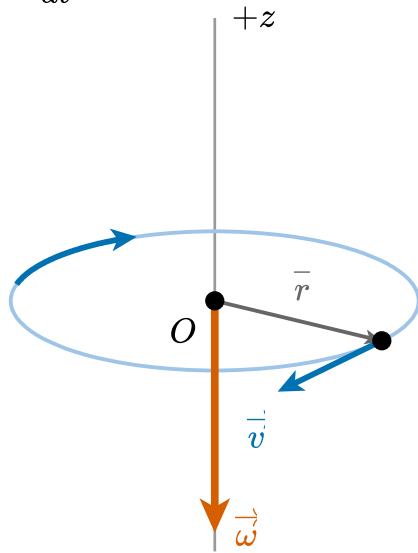


$$\omega_z = \frac{d\theta}{dt} > 0$$



counterclockwise: $\vec{\omega}$ points along $+\hat{k}$

$$\omega_z = \frac{d\theta}{dt} < 0$$



clockwise: $\vec{\omega}$ points along $-\hat{k}$